

FIZ Karlsruhe - Leibniz Institute for Information Infrastructure
AIFB - Karlsruhe Institute of Technology

4.2 Why we do need Logic

Excursion 4: A Brief Recap of Essential Logics

Excursion 5: Description Logics

4.3 First Steps in OWL

4.4 More OWL

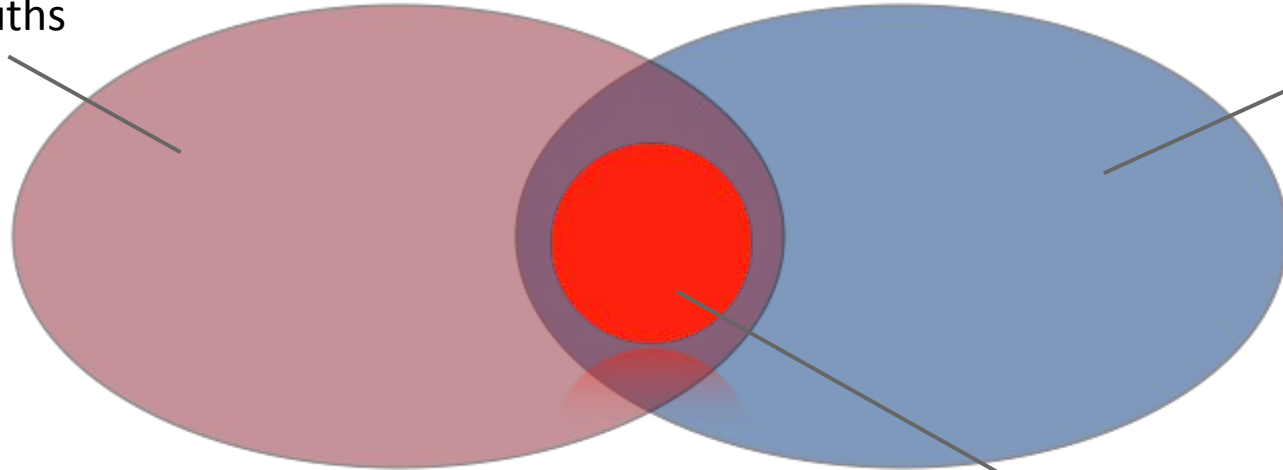
4.5 OWL and beyond

4.6 How to Design your own Ontology



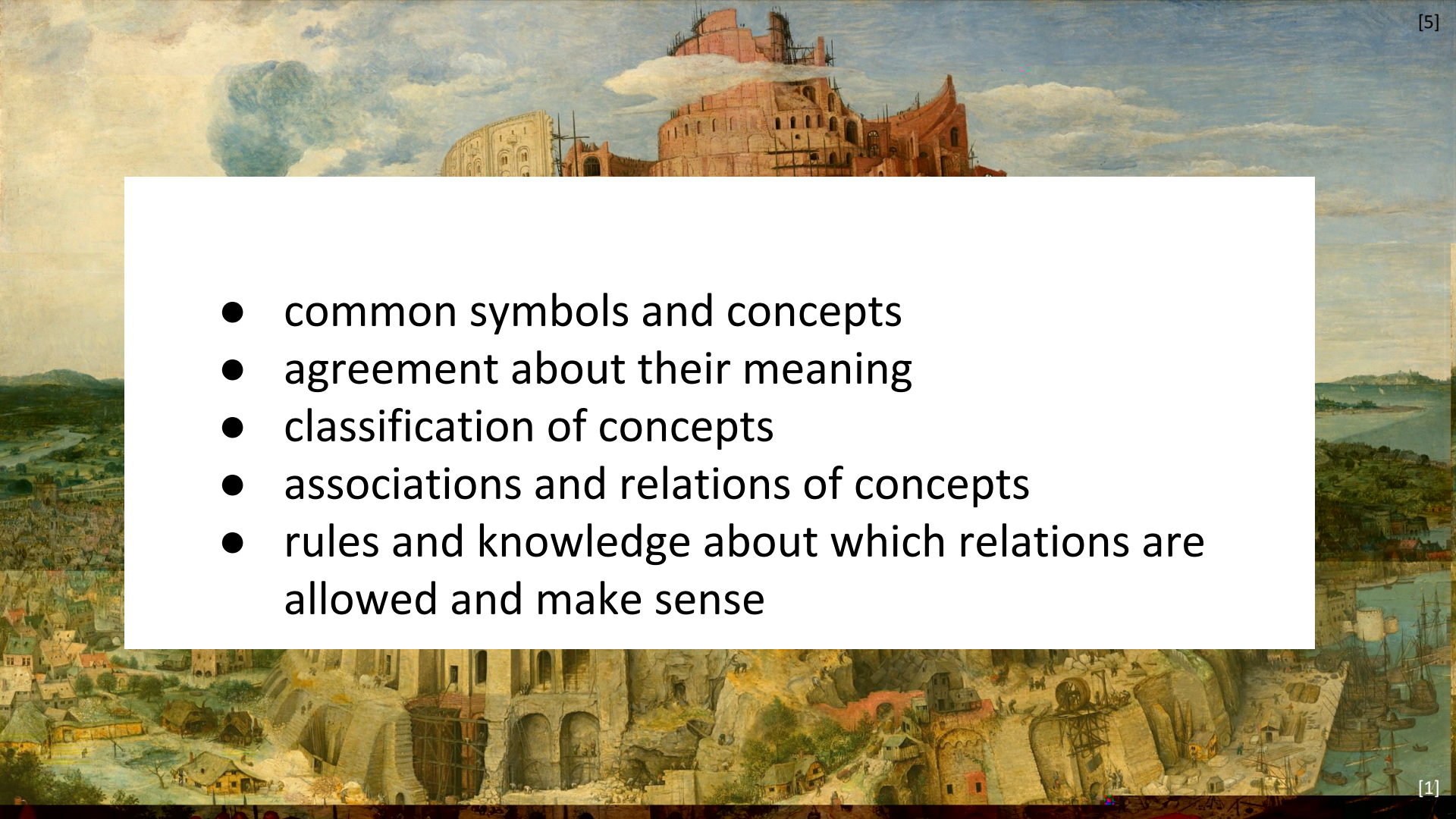
Truths

Beliefs

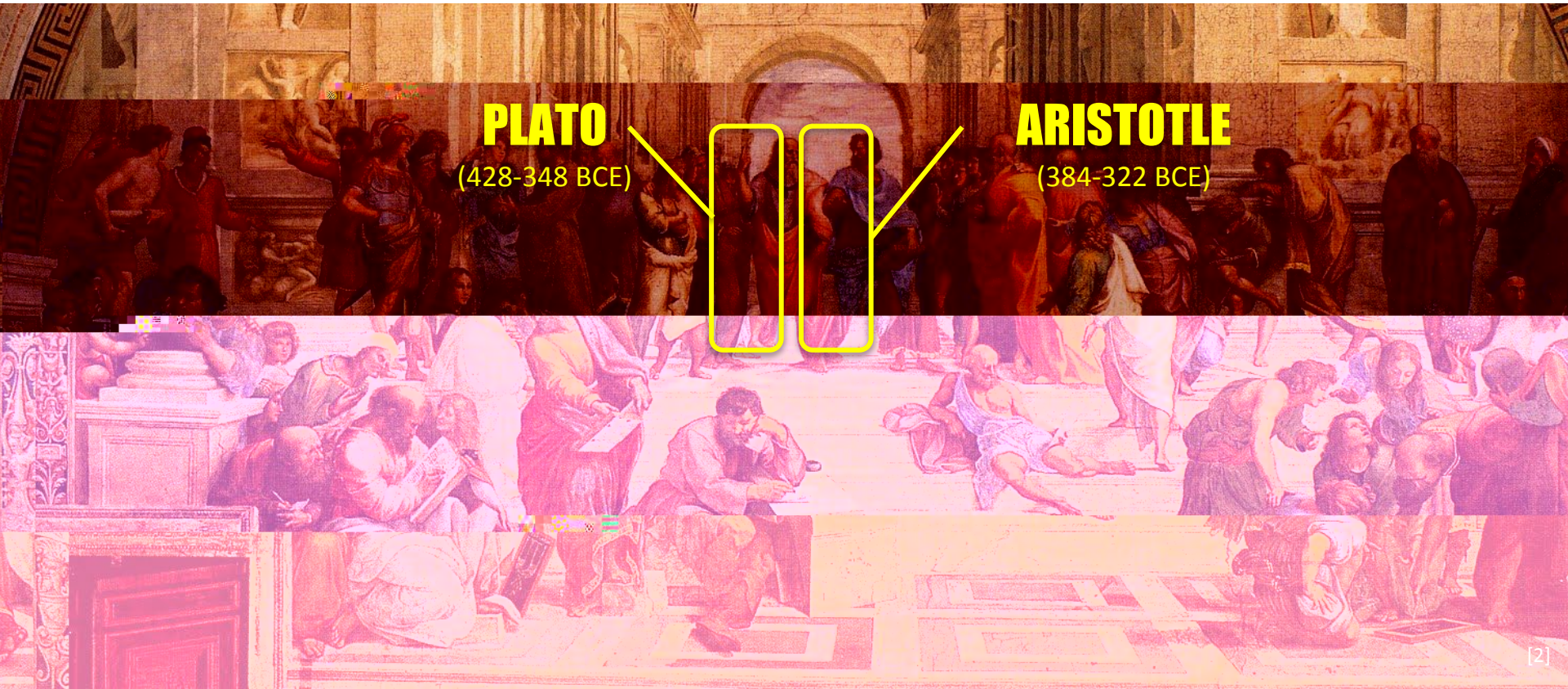


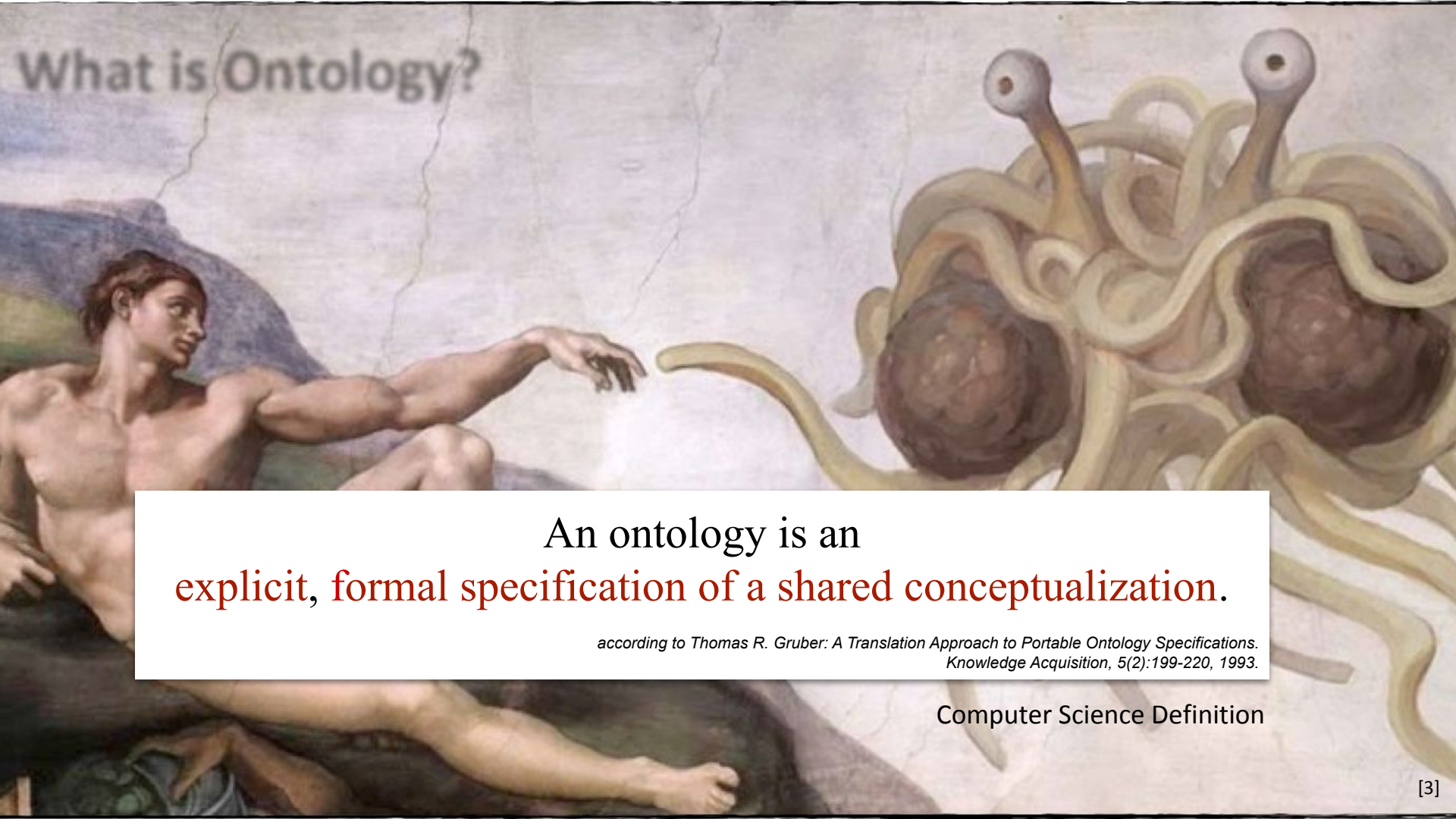
Knowledge

Traditional Definition: „Knowledge is a subset of all true beliefs.“

- 
- common symbols and concepts
 - agreement about their meaning
 - classification of concepts
 - associations and relations of concepts
 - rules and knowledge about which relations are allowed and make sense

is the philosophical study of the nature of being, existence, or reality, as well as the basic categories of being and their relations...



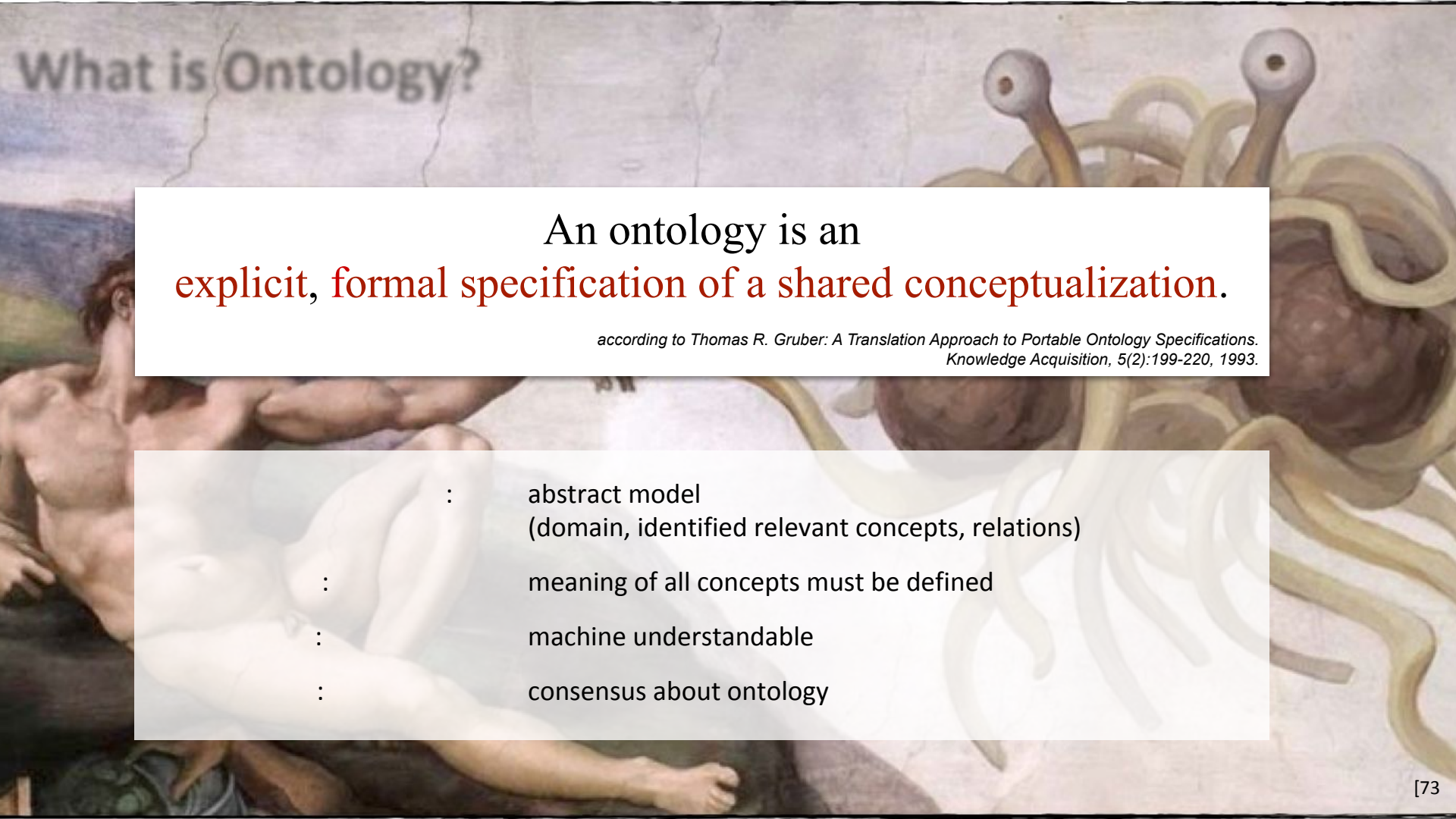


What is Ontology?

An ontology is an
explicit, formal specification of a shared conceptualization.

*according to Thomas R. Gruber: A Translation Approach to Portable Ontology Specifications.
Knowledge Acquisition, 5(2):199-220, 1993.*

Computer Science Definition



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- : abstract model
(domain, identified relevant concepts, relations)
- : meaning of all concepts must be defined
- : machine understandable
- : consensus about ontology

- Ontologies can be represented via **Classes** and **Instances**
- **Classes** are abstract groups, sets, or collections of objects and **Instances** represent
- Classes are characterised via **Properties**
- **Properties** are name-value pairs

*„**Carbon dioxide** is a colorless gas with a density about 60% higher than that of dry air. Carbon dioxide consists of a carbon atom covalently double bonded to two oxygen atoms.“*

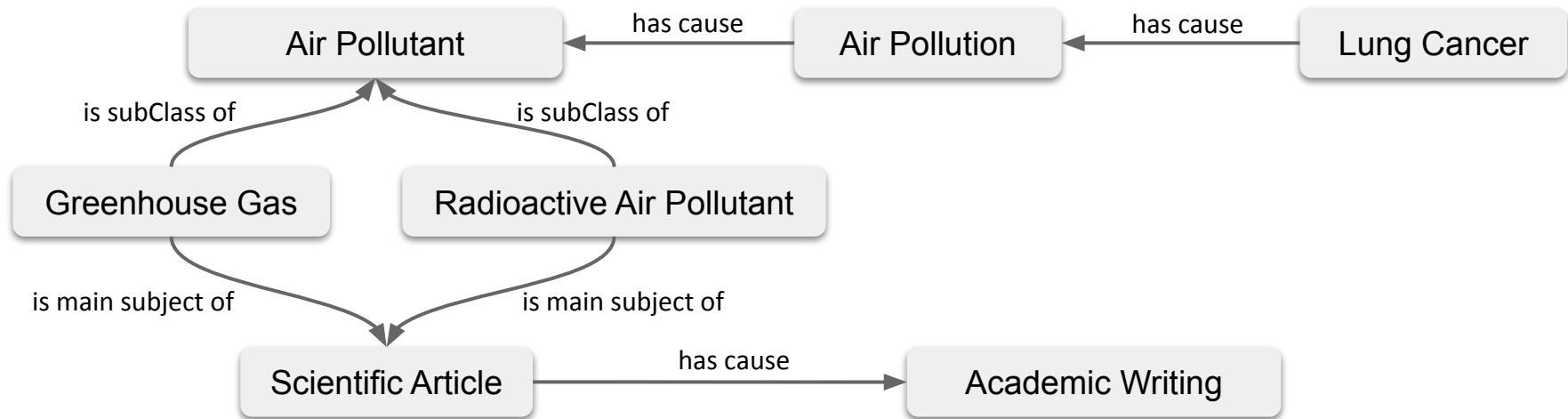
informal description

Carbon dioxide

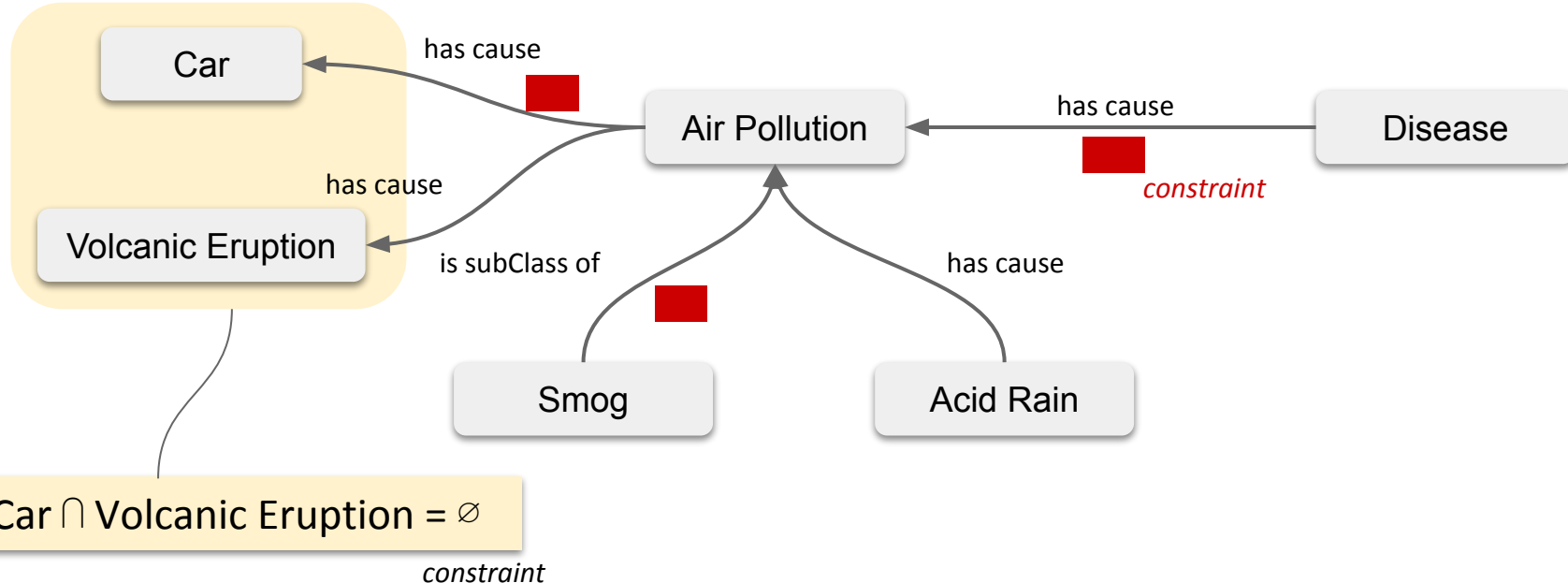
- *name <string>*
- *chemical formula <string>*
- *mass <float>*
- *density <float>*
- *sublimation temperature <string>*
- ...

semi-formal description

- Classes can be **related** to other classes
- **Properties** are special attributes, whose values are objects of (other) classes



- For Relations and Attributes () can be defined that determine allowed/valid values



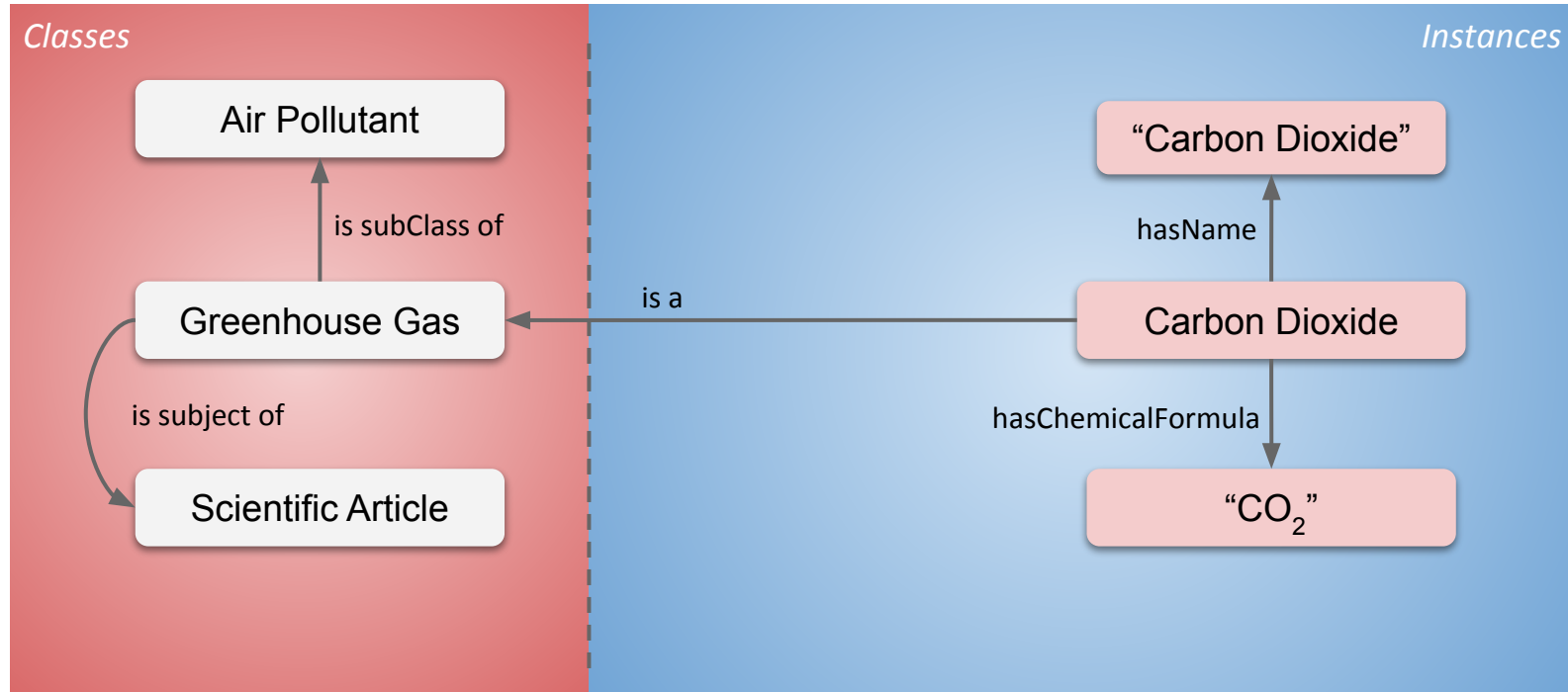
- Classes, relations, and constraints can be combined to form (complex) /
- Special Case:

Example:

- „*It is not possible to disprove climate change*“

- describe knowledge that cannot be expressed simply with the help of other existing components.

- describe individuals of an ontology





- [1] Pieter Bruegel the Elder, The Tower of Babel (1563)
[https://commons.wikimedia.org/wiki/File:Pieter Bruegel the Elder - The Tower of Babel \(Vienna\) - Google Art Project.jpg](https://commons.wikimedia.org/wiki/File:Pieter_Bruegel_the_Elder_-_The_Tower_of_Babel_(Vienna)_-_Google_Art_Project.jpg)
- [2] Raphael, School of Athens, 1511, [Public Domain]
[https://commons.wikimedia.org/wiki/File:%22The School of Athens%22 by Raffaello Sanzio da Urbino.jpg](https://commons.wikimedia.org/wiki/File:%22The_School_of_Athens%22_by_Raffaello_Sanzio_da_Urbino.jpg)
- [3] Niklas Jansson, Touched by His Noodly Appendage, the parody of Michelangelo's Creation of Adam has become an iconic image of the Flying Spaghetti Monster. 2015, [Public Domain]
[https://commons.wikimedia.org/wiki/File:Touched by His Noodly Appendage HD.jpg](https://commons.wikimedia.org/wiki/File:Touched_by_His_Noodly_Appendage_HD.jpg)
- [4] Melancholia I, Albrecht Dürer, 1511 [Public Domain]
[https://commons.wikimedia.org/wiki/File:Albrecht D%C3%9CRer - Melencolia I - Google Art Project.jpg](https://commons.wikimedia.org/wiki/File:Albrecht_D%C3%9CRer_-_Melencolia_I_-_Google_Art_Project.jpg)